PHY 111 (Spring 2025)

PRINCIPLES OF PHYSICS I

Assignment-2

Due date: 12^{th} May, 2025.

1. Two package start sliding down the 20° ramp as shown in Figure-1. Package A has a mass of 5.0 kg and coefficient of kinetic friction is 0.20. Package B has a mass of 10 kg and coefficient of kinetic friction is 0.15. Neglect the aerodynamics drag for the following calculations.

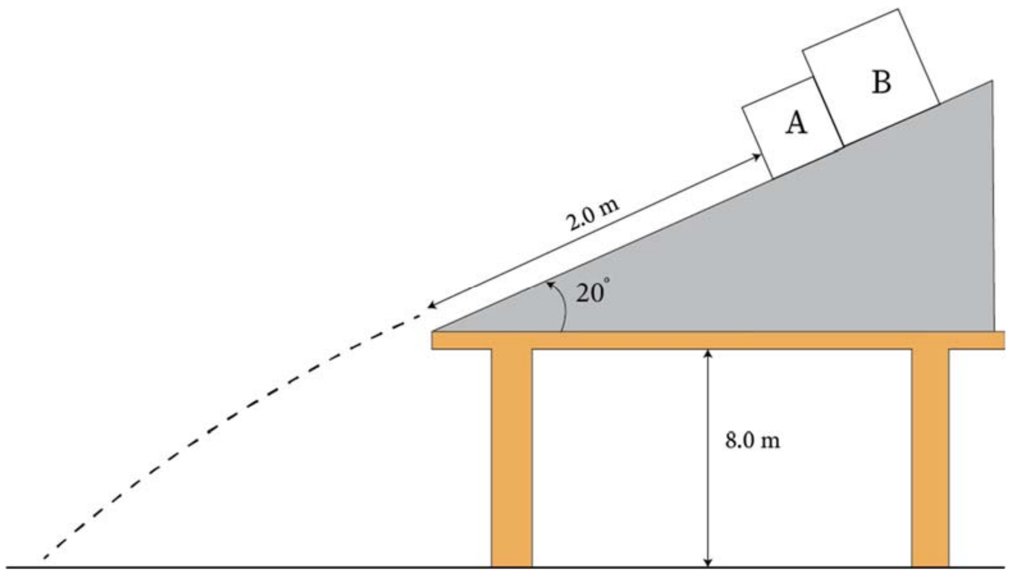

Figure-1

(i) (2 marks) Calculate the apparent weight of each Packages.

(ii) (2 mark) Calculate the maximum static frictional force on each Packages.

(iii) (6 marks) How long does the two Package will take to reach the ground starting from rest?

(iv) (3 marks) Calculate the average velocities of each Packages.

(v) (2 marks) Where does the two packages strike the ground? If we use different Package with different masses, will they strike the same point? Explain your answer.